VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belgaum-590018

Mini Project Report
on

"SURAKSHIT YATRA"

"Smart diver monitoring system and depot management system"

Second Semester of the Bachelor of Engineering Degree, towards the completion of the **Mini Project** under the **Interdisciplinary Project**
Department of Basic Sciences.
by

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CERTIFICATE

This is to certify that the File Structures mini project entitled **“SURAKSHIT YATRA -Smart driver monitoring and depot management system”** has been successfully carried out by **Akmal Haleema(1CR25CI004), Abhishek Shivakumar Komar(1CR25CI003), Pranathi(1CR25AD075), Pratik Shah(1CR25AD076)** bonafide students of **CMR Institute of Technology.**

The project is submitted in partial fulfillment of the requirements for the **Second Semester** of the Bachelor of Engineering Degree, towards the completion of the **Mini Project** under the **Interdisciplinary Project, Department of Basic Science.**

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

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Abstract:

Road accidents caused by driver fatigue and distraction are a major concern in modern transportation systems, particularly in public transport services. Continuous driving, lack of alertness, and mobile phone usage significantly increase the risk of accidents and reduce passenger safety. To address these challenges, this project presents Surakshit Yatra, an Smart Driver Monitoring and Depot Management System designed to improve road safety through real-time behavioral analysis and centralized fleet supervision.

The proposed system uses computer vision and deep learning techniques to monitor driver behavior using a standard webcam without relying on specialized hardware. MediaPipe FaceMesh is utilized for facial landmark detection to calculate Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and head posture for detecting drowsiness and yawning. YOLOv8/YOLOv10 object detection models are integrated to identify mobile phone usage and driver distractions in real-time. The system also incorporates biometric face recognition for secure driver authentication.

In addition to driver monitoring, the project includes a Smart Depot Control System that provides live bus tracking, centralized dashboard monitoring, real-time alert notifications, and automated driver assignment based on predefined schedules. Alerts are categorized into warning and danger levels, enabling immediate intervention from depot administrators during critical situations.

The developed system offers a scalable, cost-effective, and intelligent transport safety solution suitable for modern fleet management and public transportation environments. The results demonstrate successful real-time monitoring, live tracking, and efficient centralized safety management.

Chapter 1

INTRODUCTION

Road accidents caused by driver fatigue and distraction have become a major concern inmodern transportation systems. Long driving hours, lack of rest, mobile phone usage, and reduced alertness often lead to serious accidents, especially in public transport vehicles such as buses.

Traditional monitoring methods mainly rely on manual supervision and are unable to provide real-time driver safety analysis. Existing systems also depend heavily on hardware components, making them expensive and difficult to deploy on a large scale.

The objective of this project is to develop an intelligent driver monitoring and smart depot management system capable of detecting drowsiness, distraction, and unsafe driving behavior in real-time using standard webcams and software-based technologies.

The proposed system, **Surakshit Yatra**, integrates:

- Real-time facial analysis
- Drowsiness detection
- Smartphone distraction detection
- Live depot monitoring
- Fleet tracking and alert escalation

The system also includes a centralized depot dashboard where transport administrators can monitor buses, drivers, alerts, and live operational status.

The project aims to improve road safety, reduce accidents, and support smarter public transportation systems.

1.1 Brief History of Driver Monitoring Systems:

Early driver monitoring systems mainly depended on manual supervision and basic vehicle alarms. Later developments introduced hardware-based sensors such as steering movement sensors, infrared eye trackers, and biometric devices.

Modern intelligent systems now use:

- Artificial Intelligence (AI)
- Computer Vision
- Deep Learning
- Facial Landmark Detection
- Real-Time Analytics

These technologies enable accurate detection of driver fatigue and distractions without requiring complex external hardware.

1.2 Modern AI-Based Driver Safety Systems:

Modern systems use:

- Computer vision algorithms
- AI-based facial recognition
- Deep learning object detection
- Real-time analytics dashboards

Features include:

- Eye closure monitoring
- Yawning detection
- Mobile phone detection
- Live tracking
- Centralized fleet monitoring

Surakshit Yatra extends these capabilities by integrating a Smart Depot Control System for centralized safety management.

Chapter 2

PROBLEM STATEMENT

2.1 Description:

Driver fatigue and distraction are among the leading causes of road accidents in public transportation systems. Existing monitoring approaches are inefficient because:

- Drivers aremonitored manually
- No real-timealert mechanism exists
- Hardware-based systems are costly
- Fleet managers cannot monitor all vehicles centrally.

There is a need for a scalable, intelligent, and cost-effective system capable of continuously monitoring driver behavior and providing immediate intervention.

2.2 Challenge :

The major challenges addressed in this project include:

- Detecting drowsiness accurately in real-time
- Identifying mobile phone distractions
- Implementing reliable face authentication
- Managing multiple drivers efficiently
- Providing centralized depot monitoring
- Implementing live bus tracking
- Ensuring usability for drivers with low technical literacy

Chapter 3

System design Approach

3.1 Design Thinking Process:

a) Empathize

Discussions with drivers and transport staff highlighted issues such as fatigue during long journeys, delayed reactions, and lack of centralized monitoring.

b) Define

Thesystem required real-time driver monitoring with immediate alert mechanisms to improve road safety. It also needed a simple user interface along with smart depot and live fleet management features.

c) Ideate

Various approaches such as hardware-based monitoring, sensor-based systems, and AI camera-based solutions were analyzed. Finally, an software-based monitoring system using webcams and intelligent dashboards was selected.

d) Prototype

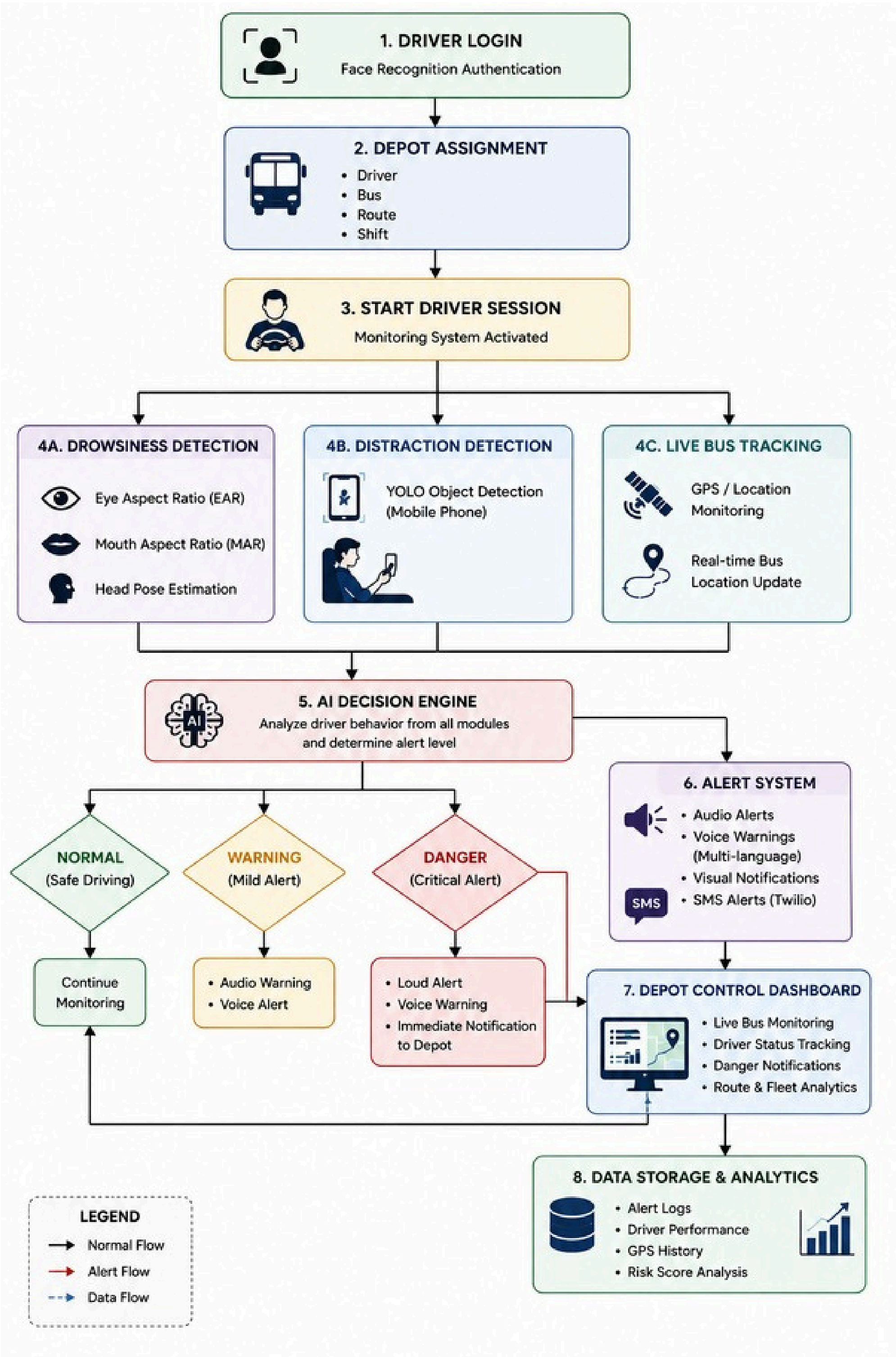
A working software prototype was developed using OpenCV, MediaPipe, YOLOv8, Flask, and face recognition modules.The prototype integrated drowsiness detection, distraction monitoring, and depot management features.

e) Test

System testing successfully demonstrated real-time drowsiness and distraction detection with accurate alert generation. The results also showed efficient depot notifications and improved overall driver monitoring performance.

3.2 Methodology

Working Procedure:

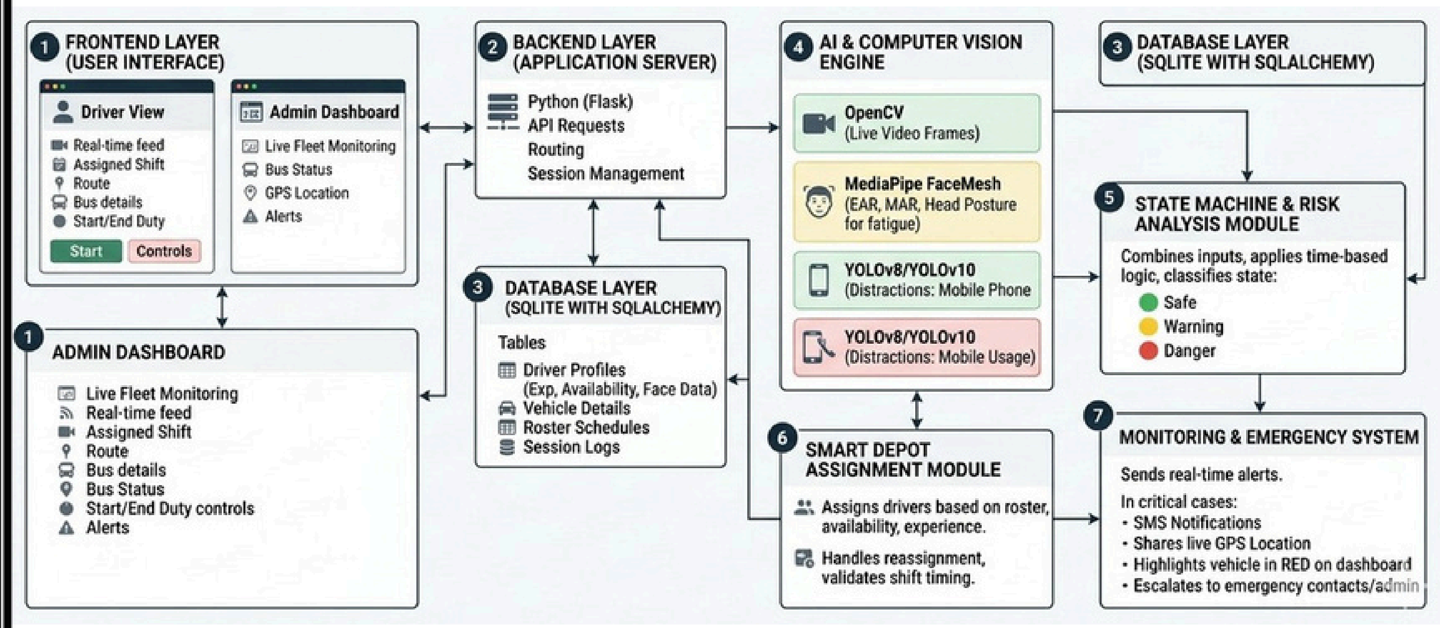


3.3 Prototype Description

3.3.1 Technologies Used

Category	Technology Used	Purpose
Web Framework	Flask	Backend development, routing, dashboard management
Authentication	Flask-Login, Werkzeug	Secure login and password hashing
Real-Time Communication	Flask-SocketIO, Socket.IO	Live GPS tracking and real-time alerts
Database	Flask-SQLAlchemy, SQLite	Data Storage and database management
Environment Configuration	python-dotenv	Managing environment variables and secrets
SMS Alerts	Pushbullet API & Telegram Bot API	Real time push notifications and Telegram alerts
Computer Vision	OpenCV	Video capture and image processing
Facial Landmark Detection	MediaPipe FaceMesh	EAR, MAR, head pose estimation
Object Detection	YOLOv8 / YOLOv10 (Ultralytics)	Mobile phone distraction detection
Face Recognition	DeepFace (FaceNet, ArcFace)	Driver biometric authentication
Deep Learning Runtime	PyTorch, TorchVision	Running YOLO deep learning models
Numerical Computation	NumPy, SciPy	Mathematical and vector calculations
Image Enhancement	Pillow (PIL)	Improving image quality for recognition
Frontend Templates	Jinja2	Dynamic HTML rendering
Styling	HTML, CSS, JavaScript	User interface design
Map Integration	Leaflet.js	Real-time live bus tracking
Voice Alerts	Web Speech API	Multilingual voice-based warnings
Camera Streaming	MJPEG Stream	Live webcam feed for monitoring

3.3.2 Software Architecture:



Chapter 4

IMPLEMENTATION

Key Code Snippets:

detector.py

```
class SafeDriveDetector:
    def __init__(self, model_path='models/yolov10n.pt'):
        self.model_path = model_path
        self.model = None

        # YOLOv10 for general detection
        if YOLO_AVAILABLE:
            try:
                self.model = YOLO(model_path)
            except Exception as e:
                print(f"Warning: Could not load YOLOv10 ({e})")
                try:
                    self.model = YOLO('yolov8n.pt')
                except Exception as e2:
                    print(f"Warning: YOLOv8 also not available ({e2})")
                    self.model = None
```

```
# YOLOv10 for general detection
if YOLO_AVAILABLE:
try:
self.model = YOLO(model_path)
except Exception as e:
print(f"Warning: Could not load YOLOv10 ({e})")
try:
self.model = YOLO('yolov8n.pt')
except Exception as e2:
print(f"Warning: YOLOv8 also not available ({e2})")
self.model = None

# MediaPipe FaceMesh for Precision Eye Tracking
self.mp_face_mesh = mp.solutions.face_mesh
self.face_mesh = None
self.face_mesh_available = False
try:
self.face_mesh = self.mp_face_mesh.FaceMesh(
max_num_faces=1,
refine_landmarks=True,
min_detection_confidence=0.5,
min_tracking_confidence=0.5
)
self.face_mesh_available = True
except Exception as e:
print(f"Warning: MediaPipe FaceMesh unavailable; running without facial landmarks ({e})")
self.mp_drawing = mp.solutions.drawing_utils
self.drawing_spec = self.mp_drawing.DrawingSpec(thickness=1, circle_radius=1, color=(0,
255, 0))

# Eye Landmark Indices (MediaPipe)
self.LEFT_EYE= [362, 382, 381, 380,374, 373, 390, 249, 263, 466, 388, 387, 386, 385, 384,
398]
self.RIGHT_EYE = [33, 7, 163, 144, 145, 153, 154, 155, 133, 173, 157, 158, 159, 160, 161,
246]
self.LEFT_IRIS = [474, 475, 476, 477]
self.RIGHT_IRIS = [469, 470, 471, 472]
self.LEFT_EYE_CORNERS = (362, 263)
self.RIGHT_EYE_CORNERS = (33, 133)
self.LEFT_EYE_VERTICAL = (386, 374)
self.RIGHT_EYE_VERTICAL = (159, 145)
```

DEPOT LIVE STATUS API

```
@app.route('/api/depot/live-status')
@login_required
def depot_live_status():
    """JSON endpoint returning real-time depot status for dashboard polling"""
    today = date.today()
#Alltoday's assignments (Assigned or In Progress)
    active_assignments = DriverAssignment.query.filter(
        DriverAssignment.assignment_date == today,
        DriverAssignment.status.in_(['Assigned', 'In Progress'])
    ).all()



    critical_alerts = Alert.query.filter_by(
        alert_type='EMERGENCY',
        severity='Critical',
        is_resolved=False
    ).all()
    emergency_drivers = {a.related_driver_id: a for a in critical_alerts}
     #Get recent general alerts for dashboard list
recent_alerts=Alert.query.filter_by(is_resolved=False).order_by(Alert.created_at.desc()).li
mit(10).all()
    recent_alerts_data = []
    for alert in recent_alerts:
        b = Bus.query.get(alert.related_bus_id) if alert.related_bus_id else None
        d = Driver.query.get(alert.related_driver_id) if alert.related_driver_id else None
        loc_str = "Status checked"
        if b and b.current_latitude is not None and b.current_longitude is not None:
            loc_str = f"Lat: {b.current_latitude:.2f}, Lng: {b.current_longitude:.2f}"
         elif d and d.last_latitude is not None and d.last_longitude is not None:
             loc_str = f"Lat: {d.last_latitude:.2f}, Lng: {d.last_longitude:.2f}"

        recent_alerts_data.append({
            'id': alert.id,
            'bus_number': b.bus_number if b else 'Unknown',
            'message': alert.message,
            'severity': alert.severity, # Critical, High, Medium, Low
            'time': alert.created_at.strftime('%I:%M %p'),
            'location': loc_str
        })
```

Chapter 5

RESULTS AND ANALYSIS

5.1 User Testing & Feedback :

Sample

Participants:

- 10 students
- 3 test drivers

5.2 Quantitative Results:

Parameter	Result
Drowsiness Detection Response	~6 seconds
Danger Alert Response	~3 seconds
Dashboard Update Delay	<2 seconds
Phone Detection	Successfully detected
Live Bus Tracking	Real-time

5.3 Qualitative Feedback:

Drivers

“The voice alerts help maintain alertness during driving.”

“The system is simple and easy to use.”

5.4 Overall Analysis:

The testing results demonstrate that Surakshit Yatra successfully performs:

- Real-time driver monitoring
- Drowsiness and distraction detection
- Face-based driver authentication
- Live bus tracking
- Smart depot alert management

5.5 Screenshots :

Figure 5.1 — Surakshit Yatra Welcome Screen

The above screen displays the welcome page of the Surakshit Yatra system. It presents the project title and official logo, providing the initial interface and branding identity of the intelligent transport safety platform.

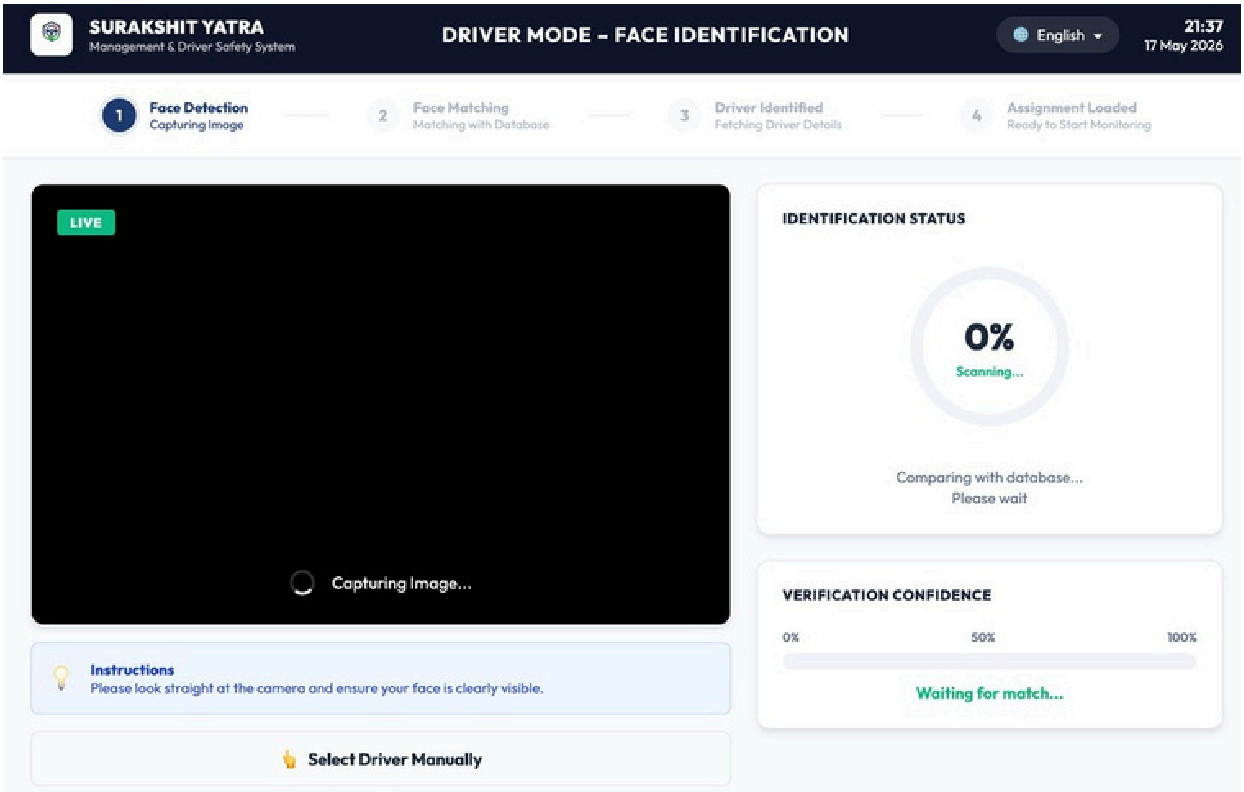


Figure 5.2 — Driver Face Authentication Page

The above screen displays the biometric login interface used for driver authentication. The system captures the driver's face through a webcam and verifies identity using AI-based face recognition before granting access to the monitoring system.

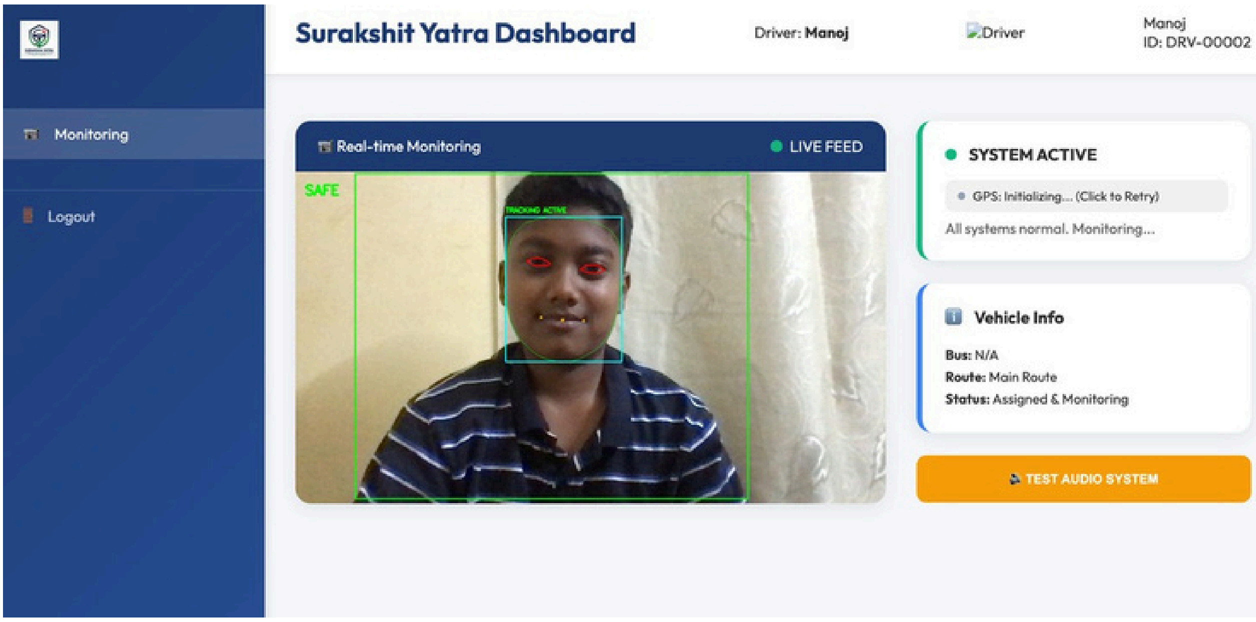


Figure 5.3 — Drowsiness Detection

The above screen demonstrates the drowsiness detection module. The system uses Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), and head posture analysis to identify fatigue symptoms such as eye closure and yawning, generating warning or danger alerts accordingly.

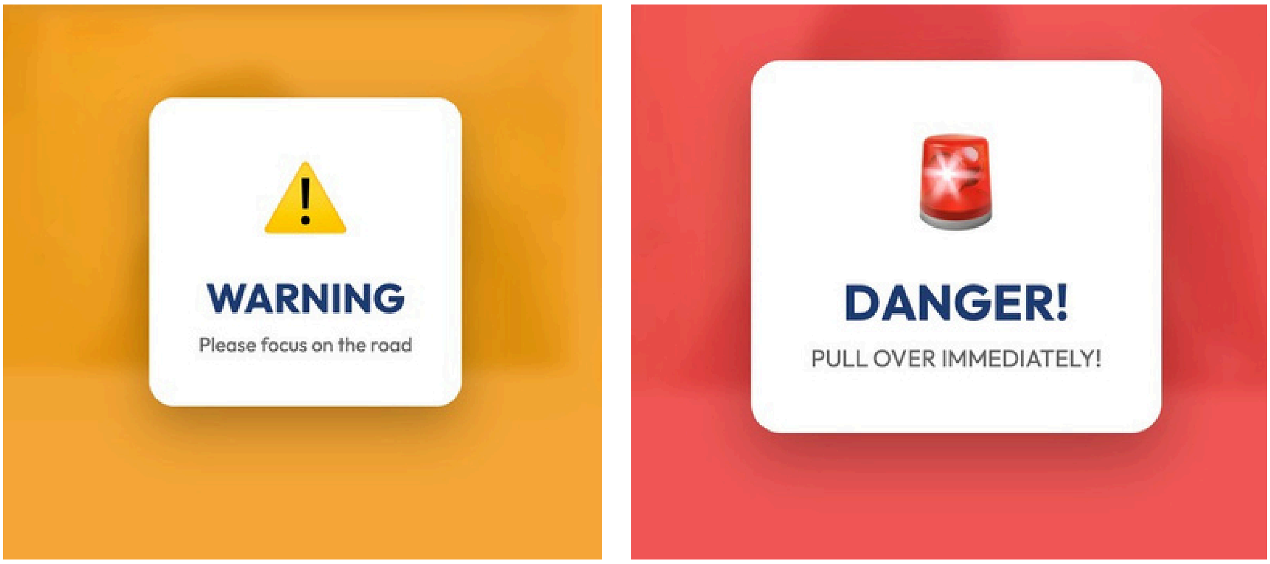


Figure 5.4 — Warning and Danger Alert System

The above screen displays the alert escalation mechanism. Depending on the severity of unsafe behavior, the system categorizes alerts into WARNING and DANGER levels and provides audio, visual, and voice notifications to the driver.

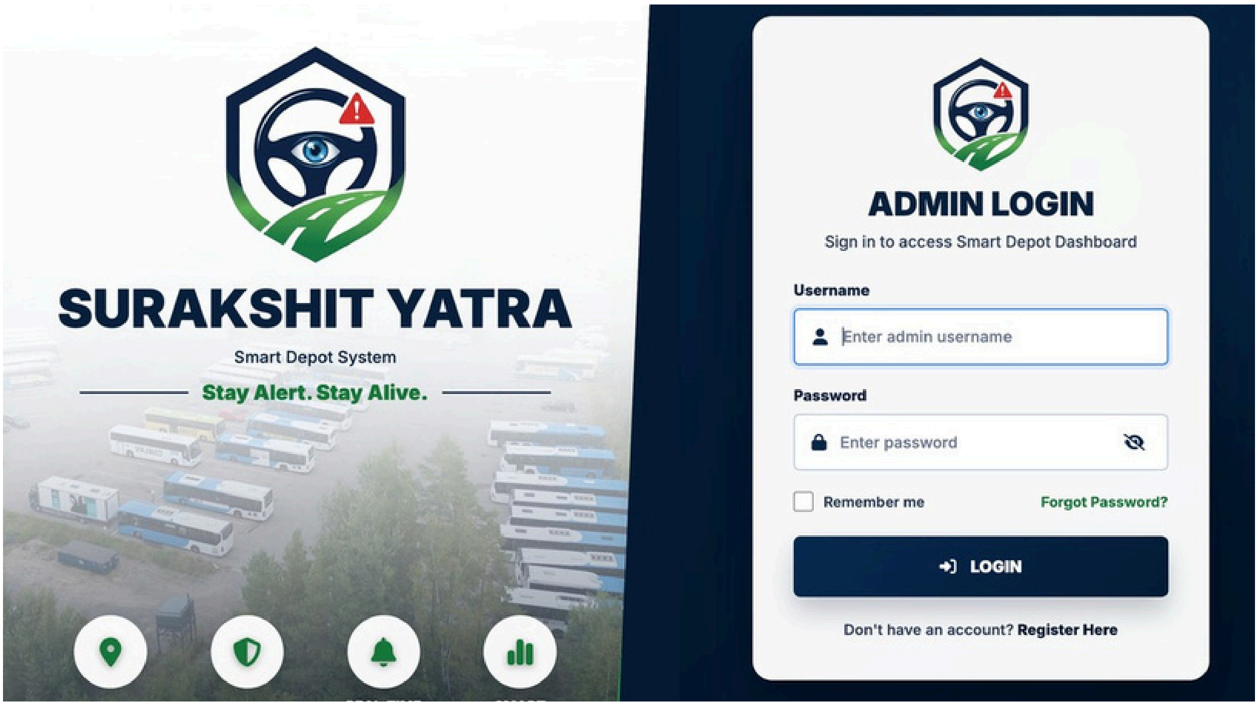


Figure 5.5 — Smart Depot Admin Login Page

The above screen displays the Smart Depot Admin Login interface. It allows authorized depot administrators to securely access the centralized dashboard for monitoring buses, drivers, alerts, live tracking, and overall fleet operations.

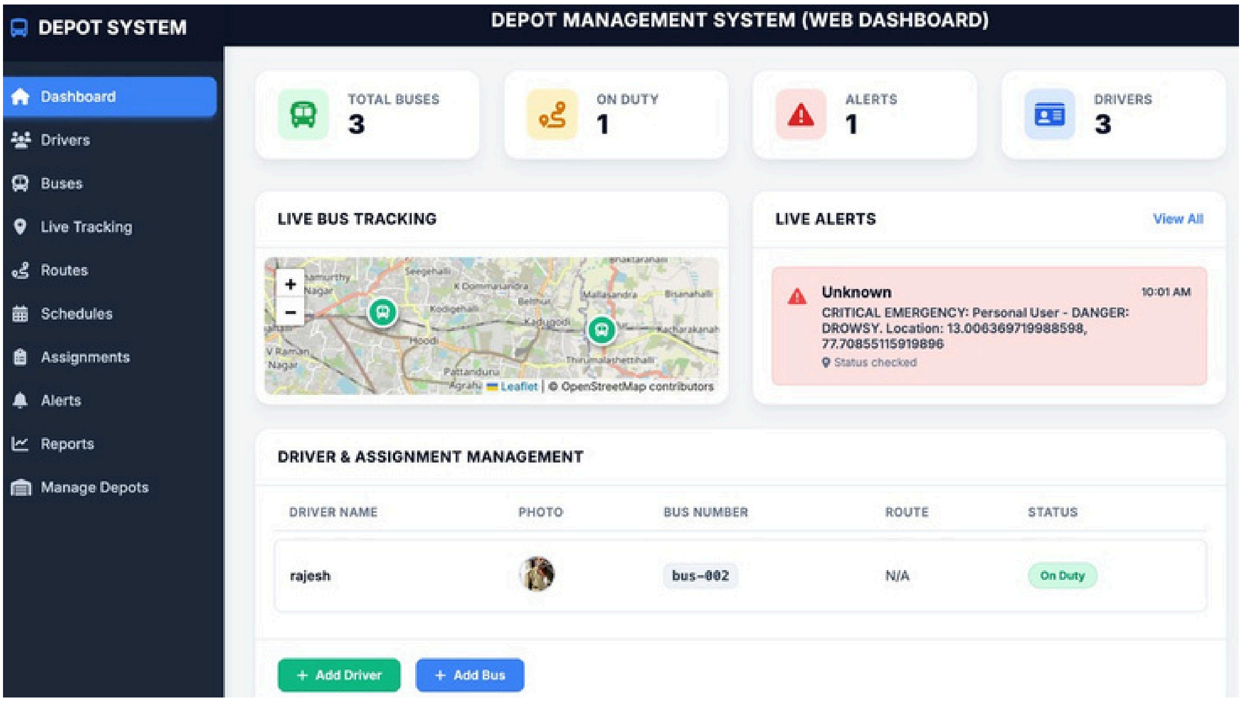


Figure 5.6— Smart Depot Dashboard with Live Bus Tracking

The above screen displays the centralized Smart Depot Dashboard integrated with live bus tracking. The dashboard enables administrators to monitor active buses, driver status, real-time vehicle locations, danger alerts, and overall fleet operations through an interactive map-based interface.

Chapter 6
Conclusion & Future Work

6.1 Conclusion:

Surakshit Yatra successfully implements an AI-powered intelligent transport safety system capable of:

- Detecting driver drowsiness
- Identifying distractions
- Managing fleet operations
- Providing centralized depot monitoring

The system improves road safety, enhances transport efficiency, and supports smarter public transportation management.

6.2 Future Enhancements:

- Advanced liveness detection
- Cloud-based fleet analytics
- GPS hardware integration
- AI-based predictive fatigue analysis
- Mobile application integration
- Government transport integration
- Multi-camera monitoring systems

References

Research References:

1.A Review of Recent Developments in Driver Drowsiness Detection Systems-
This paper reviews different driver drowsiness detection methods including behavioural, physiological, and vehicle-based techniques. It explains the advantages and limitations of each approach and highlights the importance of real-time monitoring systems for road safety.

2.Drowsiness Detection in Drivers: A Systematic Review of Deep Learning-Based Models-This study discusses how deep learning models such as CNNs and hybrid neural networks are used for driver drowsiness detection. The paper shows that vision-based deep learning systems can achieve high accuracy in real-time fatigue monitoring.

3.YOLO-CNN Powered Real-Time Detection of Visual Distractions and Drowsiness in Drivers-This paper proposes a real-time driver monitoring system using YOLO and CNN models to detect distraction and drowsiness. The system improves driver safety by providing quick and accurate detection of unsafe behaviours.

Annexures:

Annexure A – Source Code
Co ntains the Python code used for driver monitoring.

Annexure B – System Flowchart
Includes the flowchart representing the working process of Surakshit Yatra.

Annexure C – Output Screenshots
Contains screenshots of driver monitoring and alert notifications.

Annexure D – Software Requirements
- Python
- OpenCV
- TensorFlow
- NumPy

Annexure E – Test Results
Includes system quantitative result and user feedback results.

Annexure F – Research References
Contains research papers related to the project.